



Investigating the Effects of SWAP-seq Enhancer Deletions Using PRIME-Del as Validation in Coronary Artery Disease



K. Coco Xia^{1,3}, X. Shawn Cai^{2,3}, Yannick Lee-yow^{2,3}, Jesse M. Engreitz^{2,3}

¹Department of Biomedical Data Science, Stanford, CA 94305, USA. ²Department of Genetics, Stanford University School of Medicine, Stanford, CA, USA. ³Basic Science and Engineering Initiative, Betty Irene Moore Children's Heart Center, Lucile Packard Children's Hospital, Stanford University School of Medicine, Stanford, CA, USA.

Background

SWAP-seq and gene regulation by CREs

- Most of the human genome (~98%) are **non-protein coding regions**, which harbor ~90% of disease-related variants.
- Cis-regulatory elements (CREs)**, or non-coding DNA sequences, control where, when, and how much a gene is expressed.
- SWAP-seq**: a previously developed, high-throughput method to study non-coding elements.
 - Based on twin prime editing, perturbs candidate CREs individually.
 - Inserts a unique barcode cassette to ensure that edits are traceable.
 - Minimizes off-target effects compared with CRISPRi and Cas9.

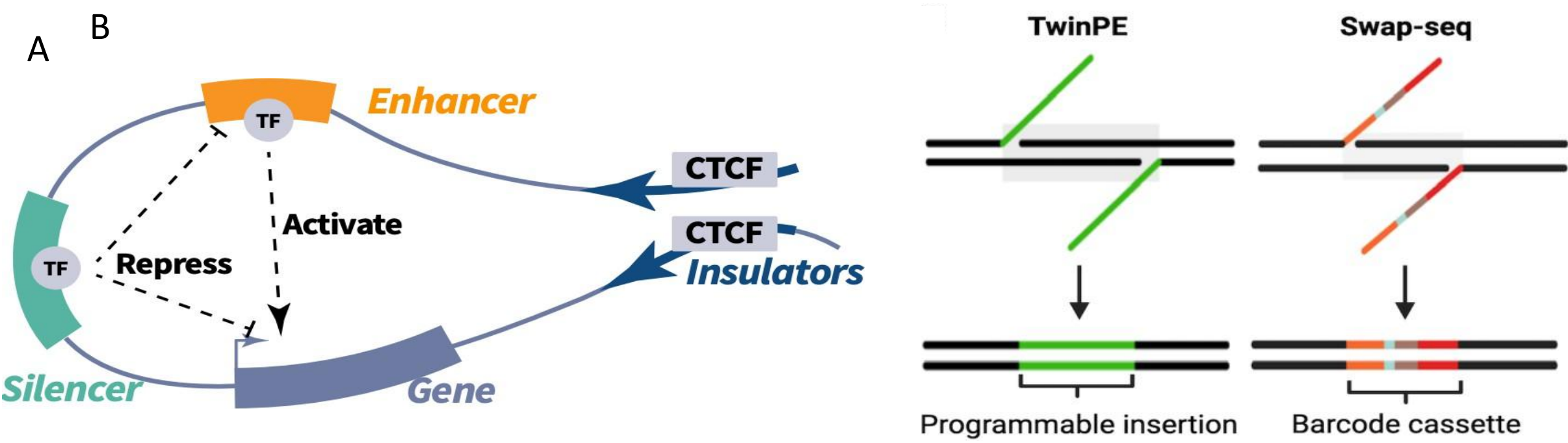


Figure 1: CREs involve enhancers, promoters, silencers, and insulators that all work together to regulate gene expression.

Figure 2: SWAP-seq is developed through the high-throughput adaptation of twinPE.

PrimeDel (PD) dissects non-coding elements

- PrimeDel (PD)**, or prime deletion, is an experimentally-validated, low-throughput application of prime editing.
- Rather than inserting a barcode cassette, PD directly perturbs the selected genetic region and concatenates the remaining region.

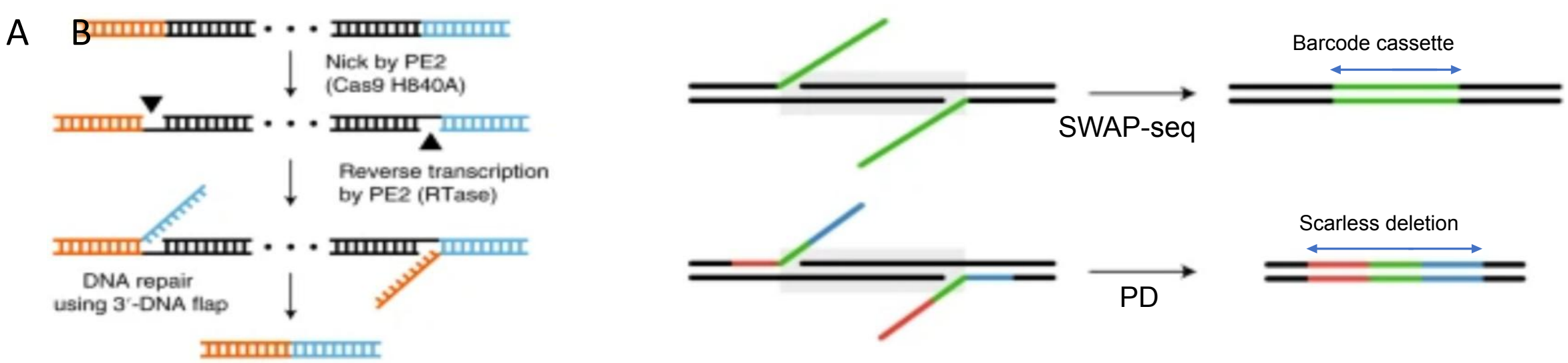


Figure 3: Different from SWAP-seq (bases on twinPE), PrimeDel (PD) introduces precise genomic deletions through a Cas9 nickase fused to a reverse transcriptase and a pegRNA.

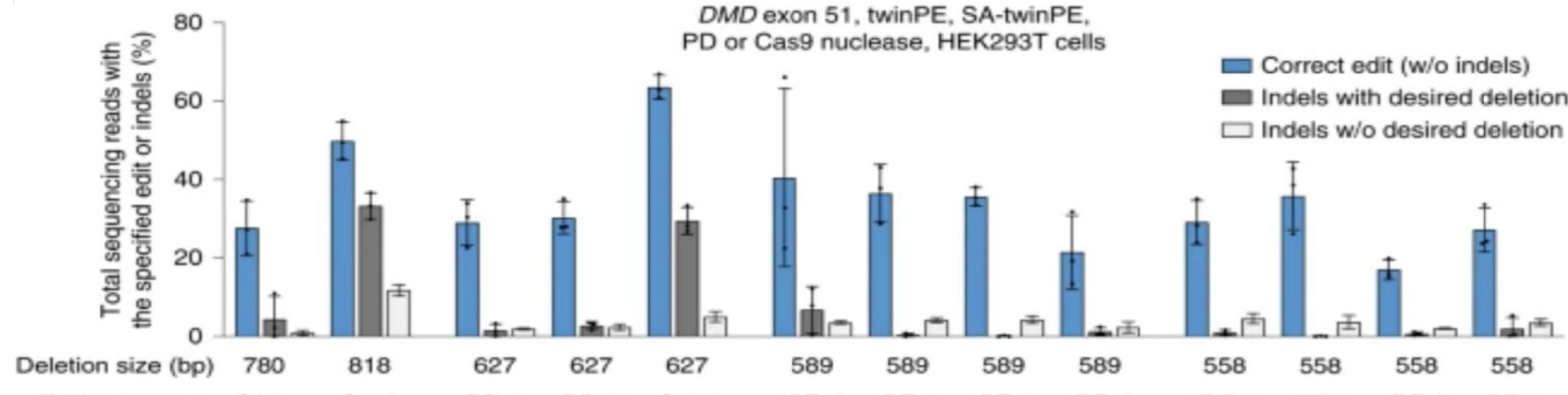


Figure 4: PD achieves higher frequencies of precise deletions with fewer indels compared to Cas9- and twinPE-based editing in HEK293T cells.

Objectives

- Apply low-throughput PrimeDel (PD) to **experimentally validate** candidate CREs identified and manipulated by SWAP-seq.
- Bridge **computational predictions** with experimental validation.
- Understand the effects of precise perturbations on gene expression, and **how non-coding elements affect gene regulation**.

Experimental Design

ePPIF enhancer as the experimental CRE

- The PPIF gene is found to have **high enhancer-gene connections**.
- Thus, disrupting multiple PPIF enhancers alters the expression of **only one gene**, which is ideal for investigating how enhancer disruption affect gene expression.

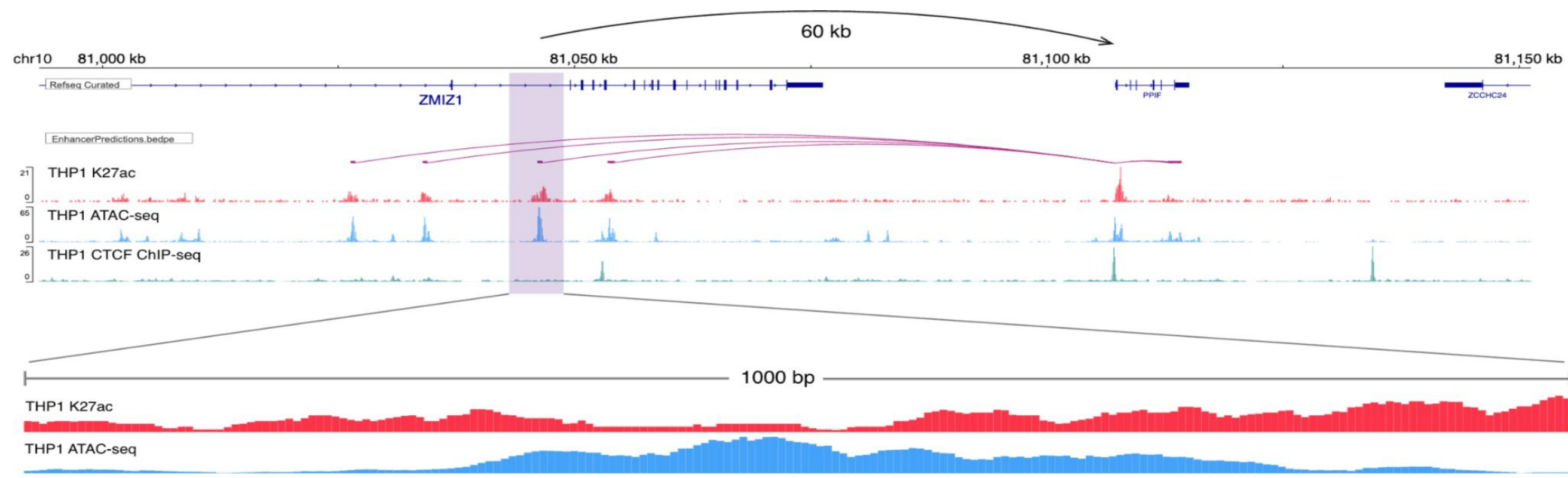


Figure 5: The PPIF enhancer, determined through key histone marks and regional accessibility.

Test pegRNA design and editing efficiency at PPIF in HEK293T Cells

- Designed three strains of pegRNAs: **pXSC109**, **pXSC110**, and **pXSC111**, each with a different length of deletion.
- Conducted three rounds of **cloning** in E. coli to generate valid plasmids using **Gibson Assembly** and **Golden Gate Assembly**.
- Transfected plasmids into **HEK293T cells** to confirm if editing is possible in THP-1 cells; tested transfection efficiency using a **PE2 prime editor fused with GFP**.

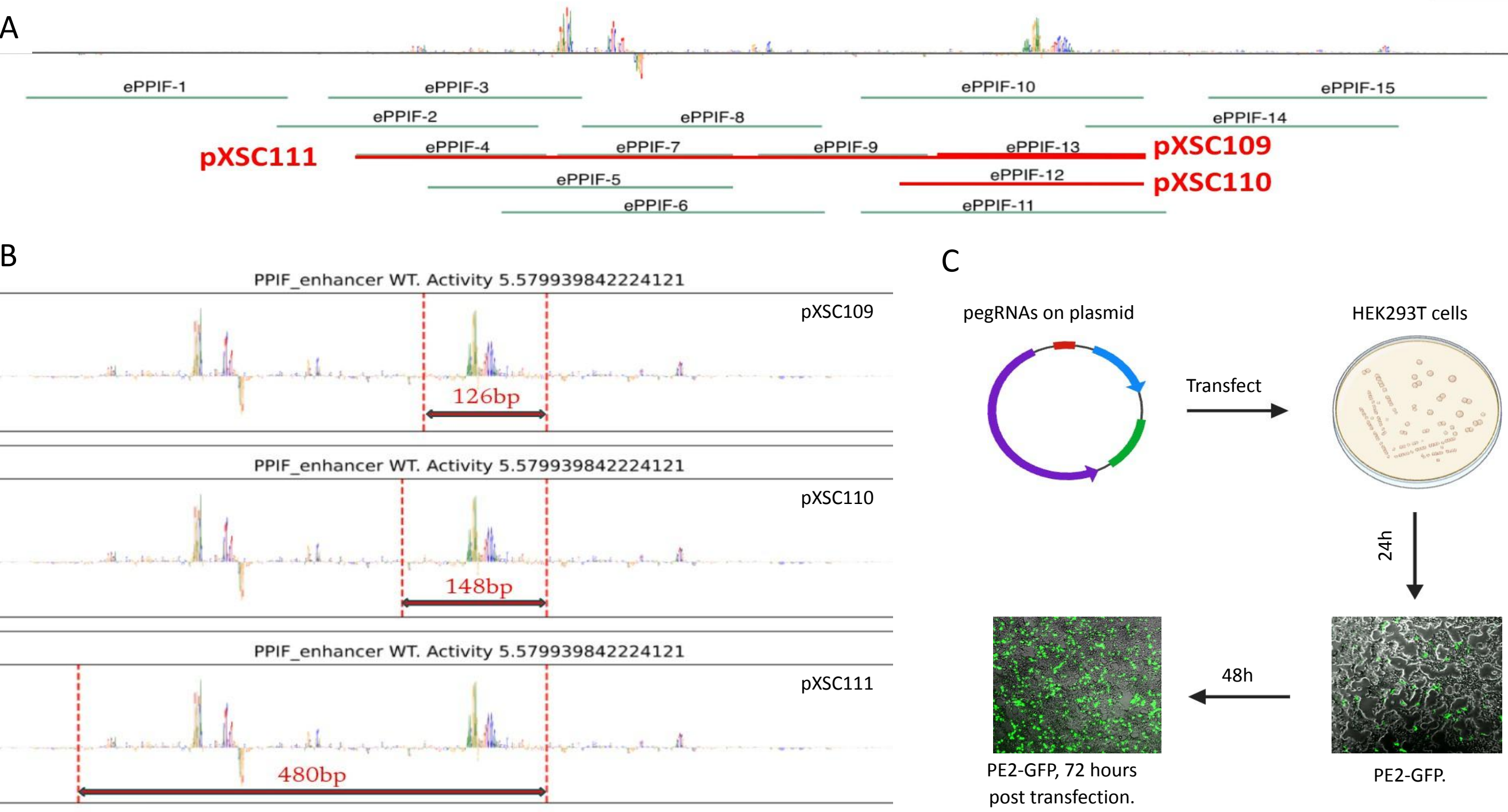
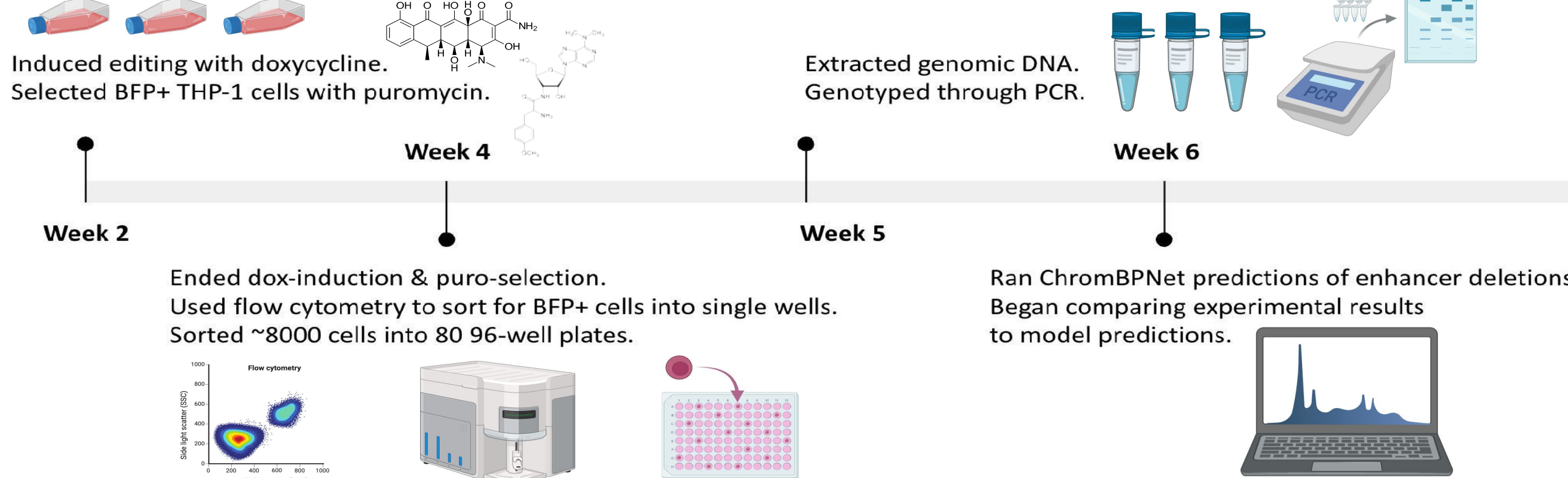


Figure 6: pegRNAs were designed and cloned into valid plasmids, then transfected into HEK293T cells to test editing viability.

PrimeDel Validation in THP-1 Cells



Results

PCR results confirm editing efficiency in HEK293T cells

- PE2 and PEMAX worked efficiently across all strains, with PE2 editing efficiency being the **strongest in pXSC109** compared to the others.
- pXSC109 is the **shortest deletion**, so it is explainable that editing efficiency is the highest because the genetic region is least perturbed.

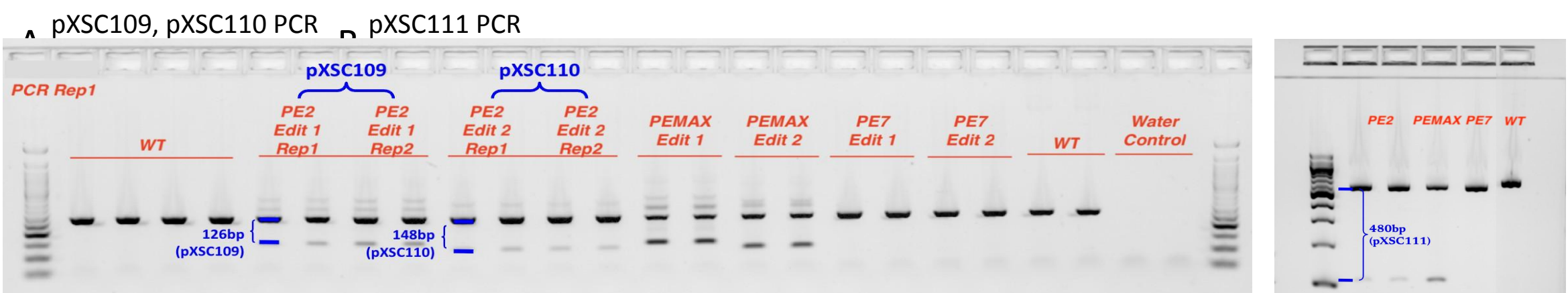


Figure 7: PCR results of pXSC109, 110, and 111 reveal successful editing.

ChromBPNet predicts edits to have varying accessibility

- ChromBPNet predictions showed no significant discrepancies between SWAP-seq and PD editing results.
- All three edits resulted in a visible decrease of gene accessibility, meaning that as CREs are perturbed, overall gene expression decreased.
- Deleting a larger region (pXSC111) resulted in a more significant drop of accessibility compared to deleting smaller regions (pXSC109, 110).

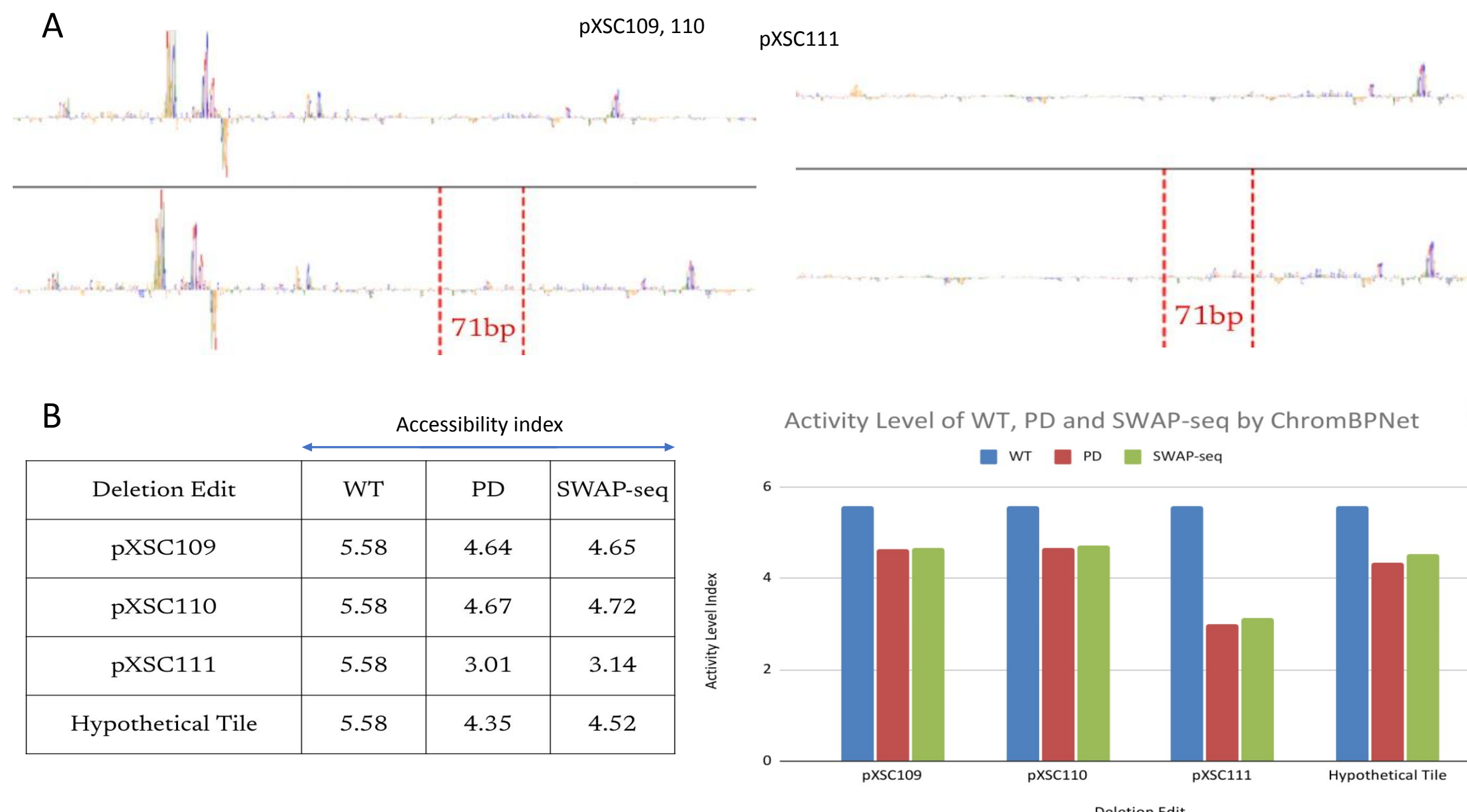


Figure 8: ChromBPNet predictions, with accessibility index comparison included, of the three edits.

Future Directions

- Finish experimentally measuring gene expression:**
 - Extract and genotype single-clone gDNA.
 - Conduct RNA extraction and RT-qPCR to directly measure gene expression.
- Validate SWAP-seq results through fluorescent reporter validation:**
 - Directly sequence selected DNA sites edited during SWAP-seq.
 - Run PCR, Gibson Assembly, transformation, and NovaGene sequencing.

References and Funding

1. Anzalone et al. (2021) Programmable deletion, replacement, integration, and inversion of large DNA sequences with twin prime editing. Nat Biotechnol.
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3. Pampari et al. (2025) ChromBPNet: bias factorized, base-resolution deep learning models of chromatin accessibility reveal cis-regulatory sequence syntax, transcription factor footprints and regulatory variants. bioRxiv.
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ENGREITZ LAB
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Department of Genetics
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